



## AQ10.2: Activity Questions 2 - Not Graded



This assignment will not be graded and is only for practice.

### Level 1:

Consider a function  $f: \mathbb{R}^2 \rightarrow \mathbb{R}$  defined as:

$$f(x, y) = \begin{cases} x^2 \sin \frac{1}{x} + y & \text{if } x \neq 0 \\ 0 & \text{if } x = 0. \end{cases}$$

- **Statement 1:**  $f_y$  is continuous at the origin.
- **Statement 2:**  $f_x$  is continuous at the origin.

Enter the number of correct statements.

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

(Type: Numeric) 0

1 point

1 point

A function  $f: \mathbb{R}^2 \rightarrow \mathbb{R}$  satisfies that  $f_x(x, y) = c$  and  $f_y(x, y) = k$  of all points of  $\mathbb{R}^2$ , where  $c$  and  $k$  are some constants, then which of the following option(s) is (are) true?

☐

The tangent line exists at any point of  $\mathbb{R}^2$  in any direction.

☐

The tangent lines at the origin do not lie on a plane.

☐

There exist points in  $\mathbb{R}^2$  for which tangent points in all directions do not exist.

☐

$f(x, y)$  can be  $f(x, y) = cx + ky + \alpha$ , where  $\alpha$  is any real number.

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

The tangent line exists at any point of  $\mathbb{R}^2$  in any direction.

$f(x, y)$  can be  $f(x, y) = cx + ky + \alpha$ , where  $\alpha$  is any real number.

1 point

Consider  $f: \mathbb{R}^2 \rightarrow \mathbb{R}$  and suppose partial derivatives of  $f(x, y)$  with respect to  $x$  and  $y$  are continuous at a point  $(a, b)$ . Then which of the following option(s) is (are) true?

☐

$$f_x(a, b) = \lim_{(x, y) \rightarrow (a, b)} f_x(x, y) = f_x(a, b)$$

☐

$$f_y(a, b) = \lim_{(x, y) \rightarrow (a, b)} f_y(x, y) = f_y(a, b)$$

☐

$$f_x(a, b) \neq \lim_{(x, y) \rightarrow (a, b)} f_x(x, y)$$

☐

$$f_y(a, b) \neq \lim_{(x, y) \rightarrow (a, b)} f_y(x, y)$$



**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

$$f_x(a, b) = \lim_{(x, y) \rightarrow (a, b)} f_x(x, y) \quad f_x(a, b) = \lim_{(x, y) \rightarrow (a, b)} f_x(x, y)$$

$$f_y(a, b) = \lim_{(x, y) \rightarrow (a, b)} f_y(x, y) \quad f_y(a, b) = \lim_{(x, y) \rightarrow (a, b)} f_y(x, y)$$

1 point

Consider a function  $f(x, y) = x^2 + y^2$ . Which of following is the tangent at the point (1,2) in the direction of (1, 1)?

☐

$$x = 1 + \frac{t}{\sqrt{2}}, y(t) = 2 + \frac{t}{\sqrt{2}}, z(t) = 5 + \frac{6t}{\sqrt{2}}$$

☐

$$x = 1 + t, y(t) = 2 + t, z(t) = 5 + 6t$$

☐

$$x = 1 + \frac{t}{\sqrt{2}}, y(t) = 2 + \frac{t}{\sqrt{2}}, z(t) = 5 + \frac{6t}{\sqrt{2}}$$

☐

$$x = 1 + t, y(t) = 2 + t, z(t) = 5 + 6t$$

☐

$$x = 1 + \frac{t}{\sqrt{2}}, y(t) = 2 + \frac{t}{\sqrt{2}}, z(t) = 5 + \frac{6t}{\sqrt{2}}$$

☐

$$x = 1 + t, y(t) = 2 + t, z(t) = 4 + \frac{6t}{\sqrt{2}}$$

☐

$$x = 1 + t, y(t) = 2 + t, z(t) = 4 + 6t$$

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

$$x = 1 + \frac{t}{\sqrt{2}}, y(t) = 2 + \frac{t}{\sqrt{2}}, z(t) = 5 + \frac{6t}{\sqrt{2}}$$

☐

$$x = 1 + t, y(t) = 2 + t, z(t) = 5 + 6t$$

☐

$$x = 1 + \frac{t}{\sqrt{2}}, y(t) = 2 + \frac{t}{\sqrt{2}}, z(t) = 5 + \frac{6t}{\sqrt{2}}$$

☐

$$x = 1 + t, y(t) = 2 + t, z(t) = 4 + 6t$$

**Level 2:**

1 point

Consider a function

$$f(x, y) = \begin{cases} \frac{x}{y} & \text{if } y \neq 0 \\ 0 & \text{if } y = 0 \end{cases} \quad f(x, y) = \begin{cases} yx & \text{if } y \neq 0 \\ 0 & \text{if } y = 0 \end{cases}$$

Which of the following is the tangent line passing through the point (1, -1) and parallel to the line

$$x(t) = \frac{t}{\sqrt{5}}, y(t) = 1 + \frac{2t}{\sqrt{5}}, z(t) = -\frac{3t}{\sqrt{5}} \quad x(t) = \quad 5$$

☐

$$t, y(t) = 1 + \quad 5$$

☐

$$2t, z(t) = - \quad 5$$

☐

$$3t?$$

☐

$$x(t) = 1 + t, y(t) = -1 + 2t, z(t) = -\frac{3}{\sqrt{5}}t \quad x(t) = 1 + t, y(t) = -1 + 2t, z(t) = - \quad 5$$

☐

$$3t$$

☐

$$x(t) = 1 + t, y(t) = 1 + 2t, z(t) = -1 - \frac{3}{\sqrt{5}}t \quad x(t) = 1 + t, y(t) = 1 + 2t, z(t) = -1 - \quad 5$$

☐

$$3t$$

☐

$$x(t) = 1 + \frac{t}{\sqrt{5}}, y(t) = -1 + \frac{2t}{\sqrt{5}}, z(t) = -1 - \frac{3t}{\sqrt{5}} \quad x(t) = 1 + \quad 5$$

☐

$$t, y(t) = -1 + \quad 5$$

☐

$$2t, z(t) = -1 - \quad 5$$

☐

$$3t$$

☐

$$x(t) = -1 + t, y(t) = -1 + 2t, z(t) = -1 - \frac{3}{\sqrt{5}}t \quad x(t) = -1 + t, y(t) = -1 + 2t, z(t) = -1 - \quad 5$$

☐

$$3t$$

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

$$x(t) = 1 + \frac{t}{\sqrt{5}}, y(t) = -1 + \frac{2t}{\sqrt{5}}, z(t) = -1 - \frac{3t}{\sqrt{5}} \quad x(t) = 1 + \quad 5$$

☐

$$t, y(t) = -1 + \quad 5$$

☐

$$2t, z(t) = -1 - \quad 5$$

☐

$$3t$$

*1 point*

Consider a function  $f(x, y, z) = 2x^2 + xy + z^2$   $f(x, y, z) = 2x^2 + xy + z^2$ . Which of following is the tangent at the point (1,1,1) in the direction of (1, 0, 0)

☐

$$(1, 1, 1, 4 + 5t)(1, 1, 1, 4 + 5t)$$

☐

$$(1 + t, 1, 1, 4 + t)(1 + t, 1, 1, 4 + t)$$

☐

$$(1, 1 + t, 1 + t, 4 + 5t)(1, 1 + t, 1 + t, 4 + 5t)$$

☐

$$(1 + t, 1, 1, 4 + 5t)(1 + t, 1, 1, 4 + 5t)$$

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

$$(1 + t, 1, 1, 4 + 5t)(1 + t, 1, 1, 4 + 5t)$$

1 point

Consider the following functions  $f$  defined as:

$$f(x, y, z) = xyz. f(x, y, z) = xyz.$$

Then the tangent line at the point  $(1,0,1)$  in the direction of  $(1,1,1)$  is parallel to the line

☐

$$x(t) = 1 + \frac{t}{\sqrt{3}}, y(t) = \frac{t}{\sqrt{3}}, z(t) = 1 + \frac{t}{\sqrt{3}}, u(t) = \frac{t}{\sqrt{3}} x(t) = 1 + \frac{t}{\sqrt{3}}$$

☐

$$x(t) = 1 + \frac{t}{\sqrt{3}}, y(t) = \frac{t}{\sqrt{3}}, z(t) = 1 + \frac{t}{\sqrt{3}}, u(t) = \frac{t}{\sqrt{3}} x(t) = 1 + \frac{t}{\sqrt{3}}$$

☐

$$x(t) = 1 + \frac{t}{\sqrt{3}}, y(t) = \frac{t}{\sqrt{3}}, z(t) = 1 + \frac{t}{\sqrt{3}}, u(t) = \frac{t}{\sqrt{3}} x(t) = 1 + \frac{t}{\sqrt{3}}$$

☐

$$x(t) = 1 + \frac{t}{\sqrt{3}}, y(t) = \frac{t}{\sqrt{3}}, z(t) = 1 + \frac{t}{\sqrt{3}}, u(t) = \frac{t}{\sqrt{3}} x(t) = 1 + \frac{t}{\sqrt{3}}$$

☐

t

☐

$$x(t) = 1 + \frac{t}{\sqrt{3}}, y(t) = \frac{t}{\sqrt{3}}, z(t) = 1 + \frac{t}{\sqrt{3}}, u(t) = t\sqrt{3} x(t) = 1 + \frac{t}{\sqrt{3}}$$

☐

$$x(t) = 1 + \frac{t}{\sqrt{3}}, y(t) = \frac{t}{\sqrt{3}}, z(t) = 1 + \frac{t}{\sqrt{3}}, u(t) = \frac{t}{\sqrt{3}} x(t) = 1 + \frac{t}{\sqrt{3}}$$

☐

$$x(t) = 1 + \frac{t}{\sqrt{3}}, y(t) = \frac{t}{\sqrt{3}}, z(t) = 1 + \frac{t}{\sqrt{3}}, u(t) = \frac{t}{\sqrt{3}} x(t) = 1 + \frac{t}{\sqrt{3}}$$

☐

$$x(t) = 1 + \frac{t}{\sqrt{3}}, y(t) = \frac{t}{\sqrt{3}}, z(t) = 1 + \frac{t}{\sqrt{3}}, u(t) = \frac{t}{\sqrt{3}} x(t) = 1 + \frac{t}{\sqrt{3}}$$

☐

☐

$$x(t) = 1 + \frac{t}{\sqrt{3}}, y(t) = 1 + \frac{t}{\sqrt{3}}, z(t) = 1 + \frac{t}{\sqrt{3}}, u(t) = \frac{t}{\sqrt{3}} x(t) = 1 + \frac{t}{\sqrt{3}}$$

☐

$$x(t) = 1 + \frac{t}{\sqrt{3}}, y(t) = 1 + \frac{t}{\sqrt{3}}, z(t) = 1 + \frac{t}{\sqrt{3}}, u(t) = \frac{t}{\sqrt{3}} x(t) = 1 + \frac{t}{\sqrt{3}}$$

☐

$$x(t) = 1 + \frac{t}{\sqrt{3}}, y(t) = 1 + \frac{t}{\sqrt{3}}, z(t) = 1 + \frac{t}{\sqrt{3}}, u(t) = \frac{t}{\sqrt{3}} x(t) = 1 + \frac{t}{\sqrt{3}}$$

☐

$$x(t) = 1 + \frac{t}{\sqrt{3}}, y(t) = 1 + \frac{t}{\sqrt{3}}, z(t) = 1 + \frac{t}{\sqrt{3}}, u(t) = \frac{t}{\sqrt{3}} x(t) = 1 + \frac{t}{\sqrt{3}}$$

☐

t

☐ None of the above.

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

$$x(t) = 1 + \frac{t}{\sqrt{3}}, y(t) = \frac{t}{\sqrt{3}}, z(t) = 1 + \frac{t}{\sqrt{3}}, u(t) = \frac{t}{\sqrt{3}} x(t) = 1 + \frac{t}{\sqrt{3}}$$

$$\sqrt{t}, y(t) = 3$$

$$\sqrt{t}, z(t) = 1 + 3$$

$$\sqrt{t}, u(t) = 3$$

$$\sqrt{t}$$

[Check Answers](#)

Your score is: 0/7